

# Uncertainty Quantification for Cancer Drug Response Prediction Using Graph Neural Networks and Conformal Prediction

## Abstract

Predicting cancer drug response is a critical challenge in precision oncology, where accurate predictions can guide treatment selection and improve patient outcomes. While deep learning models have demonstrated strong predictive performance on drug sensitivity data, most approaches provide only point estimates without quantifying the uncertainty inherent in their predictions. This limitation hinders clinical applicability, as practitioners require reliable confidence measures to make informed decisions. This work presents a framework that combines Graph Neural Networks (GNNs) with conformal prediction to generate prediction intervals with rigorous statistical coverage guarantees for cancer drug response prediction. The proposed approach encodes drug molecules as graphs using Graph Convolutional Networks and processes cell line gene expression profiles through multilayer perceptrons, fusing these representations to predict log IC<sub>50</sub> values. Split conformal prediction is then applied to transform point predictions into prediction intervals that achieve valid coverage without distributional assumptions. Experiments on the Genomics of Drug Sensitivity in Cancer (GDSC1) dataset, comprising 177,310 drug response measurements across 208 compounds and 958 cell lines, demonstrate that the model achieves a Pearson correlation of 0.909 and an RMSE of 1.184 on held-out test data. The conformal prediction intervals achieve 90.2% empirical coverage against a 90% target, with a mean interval width of 3.795 log IC<sub>50</sub> units. Comparative analysis with Monte Carlo dropout reveals that conformal prediction provides superior coverage calibration while maintaining computational efficiency. The framework offers a principled approach to uncertainty aware drug response prediction that could enhance the reliability of computational tools in drug discovery pipelines.

## 1. Introduction

Cancer remains one of the leading causes of mortality worldwide, and the development of effective therapeutic strategies is complicated by the heterogeneous nature of the disease [1]. Different tumors, even those arising from the same tissue type, can exhibit vastly different responses to the same drug due to variations in their molecular profiles. This observation has motivated the shift toward precision oncology, where treatment decisions are guided by the molecular characteristics of individual tumors rather than population level statistics [2]. Central to this paradigm is the ability to predict how a specific cancer cell line or tumor will respond to a given drug based on genomic features.

The prediction of drug response, typically measured through metrics such as the half maximal inhibitory concentration (IC50), has been approached using various machine learning and deep learning techniques [3]. Recent advances have leveraged the structural information of drug molecules through graph-based representations, enabling models to learn directly from molecular topology rather than relying on handcrafted fingerprints [4]. Graph Neural Networks have emerged as particularly effective architectures for this task, as they can capture the complex relationships between atoms and bonds that determine molecular properties [5].

Despite the impressive predictive accuracy achieved by modern deep learning models, a critical limitation persists in their clinical translation. Most models produce point estimates without any indication of prediction confidence or uncertainty. In a clinical context, a prediction of IC50 equal to 2.5 micromolar is far more actionable if accompanied by a confidence interval indicating whether the true value is likely to fall between 2.0 and 3.0 or between 0.5 and 4.5. Without uncertainty quantification, practitioners cannot distinguish between predictions the model makes with high confidence and those where the model is essentially guessing.

Conformal prediction offers an elegant solution to this problem by providing a distribution-free framework for constructing prediction intervals with finite sample coverage guarantees [6]. Unlike Bayesian approaches that require specifying prior distributions or heuristic methods that provide uncalibrated uncertainty estimates, conformal prediction guarantees that prediction intervals will contain the true value with a user specified probability, regardless of the underlying data distribution. This property makes it particularly attractive for high stakes applications where reliable uncertainty quantification is essential.

This work makes several contributions to the field of computational drug response prediction. First, it presents an end-to-end framework that integrates GNN based drug response prediction with split conformal prediction for uncertainty quantification. Next, it provides a comprehensive empirical evaluation comparing conformal prediction with Monte Carlo dropout, a popular alternative uncertainty quantification method. Moreover, it analyzes the conditional coverage properties of conformal prediction across different prediction ranges, revealing important insights about where the method performs well and where it may require further refinement. Lastly, it demonstrates the practical utility of the framework on a large-scale dataset, showing that rigorous uncertainty quantification can be achieved without sacrificing predictive performance.

## **2. Related Work**

### **2.1 Drug Response Prediction**

The computational prediction of drug response has evolved significantly over the past decade. Early approaches relied on linear models applied to gene expression signatures or drug fingerprints [7]. The Cancer Cell Line Encyclopedia (CCLE) and Genomics of Drug Sensitivity

in Cancer (GDSC) projects provided large-scale datasets that enabled the development of more sophisticated models [8, 9]. Subsequent work explored random forests, support vector machines, and various ensemble methods, with performance improvements driven largely by feature engineering and data preprocessing [10].

The application of deep learning to drug response prediction began with fully connected networks operating on concatenated drug and cell line features [11]. These models demonstrated that neural networks could capture nonlinear relationships between molecular features and drug sensitivity. However, the representation of drugs as fixed length fingerprint vectors limited the ability of these models to leverage the full structural information encoded in molecular graphs.

## **2.2 Graph Neural Networks for Molecular Property Prediction**

Graph Neural Networks have revolutionized molecular property prediction by enabling end-to-end learning from molecular structures [5]. The message passing neural network framework provides a general paradigm where node features are iteratively updated by aggregating information from neighbouring nodes [12]. Graph Convolutional Networks, Graph Attention Networks, and related architectures have been successfully applied to tasks ranging from solubility prediction to binding affinity estimation [13, 14].

In the context of drug response prediction, GNNs have been combined with various representations of cell line features. The DeepCDR model used graph convolutions for drugs and autoencoders for genomic features [15]. GraphDRP explored different graph architectures and demonstrated improvements over fingerprint-based methods [16]. More recent work has incorporated attention mechanisms and transformer architectures to capture drug cell line interactions [17].

## **2.3 Uncertainty Quantification in Deep Learning**

Uncertainty quantification in neural networks has been approached from several perspectives. Bayesian neural networks place distributions over network weights and marginalize over the posterior to obtain predictive distributions [18]. While theoretically principled, exact Bayesian inference is intractable for large networks, necessitating approximations such as variational inference or Monte Carlo sampling.

Monte Carlo dropout provides a practical approximation to Bayesian inference by interpreting dropout at test time as approximate variational inference [19]. By performing multiple forward passes with dropout enabled, one obtains a distribution of predictions whose variance serves as an uncertainty estimate. This approach is computationally efficient as it requires only a single trained model with dropout layers, making it attractive for large-scale applications.

However, these methods provide only heuristic uncertainty estimates without formal coverage guarantees. The prediction intervals derived from dropout uncertainty may systematically

undercover or overcover depending on the dataset and model architecture, limiting their reliability in critical applications.

## 2.4 Conformal Prediction

Conformal prediction, introduced by Vovk and colleagues, provides a framework for constructing prediction sets with guaranteed coverage [6, 20]. The key insight is that if the calibration and test data are exchangeable, one can use the empirical distribution of conformity scores on a held-out calibration set to construct prediction intervals for new test points.

Split conformal prediction simplifies the full conformal procedure by using a single calibration set rather than retraining the model for each test point [21]. This approach trades some statistical efficiency for computational tractability, making it practical for large-scale applications. Recent work has extended conformal prediction to various settings including classification, structured prediction, and distribution shift [22, 23].

The application of conformal prediction to drug discovery and molecular property prediction remains relatively unexplored. Some work has applied conformal prediction to QSAR models, but comprehensive evaluations combining GNNs with conformal prediction for drug response prediction are lacking [24]. This work addresses this gap by providing a thorough empirical study of conformal prediction in this domain.

## 3. Methodology

### 3.1 Problem Formulation

The drug response prediction task can be formulated as a regression problem. Given a drug molecule represented by its molecular structure and a cancer cell line characterized by its gene expression profile, the goal is to predict the drug sensitivity, measured as the logarithm of the IC50 value. Lower log IC50 values indicate higher sensitivity, meaning the drug is effective at lower concentrations.

Formally, let  $d$  denote a drug molecule and  $c$  denote a cell line. The drug is represented as a molecular graph  $G_d = (V, E, X)$  where  $V$  is the set of atoms,  $E$  is the set of bonds, and  $X \in \mathbb{R}^{|V| \times f}$  contains atom features. The cell line is represented by its gene expression vector  $\mathbf{g}_c \in \mathbb{R}^{n_g}$  where  $n_g$  is the number of genes. The target is the log IC50 value  $y \in \mathbb{R}$ .

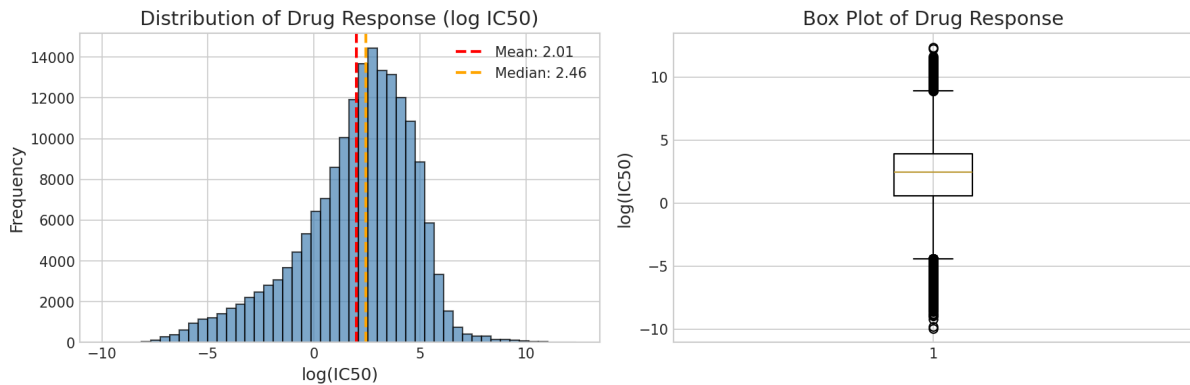
### 3.2 Dataset

This work utilizes the GDSC1 dataset from the Therapeutics Data Commons [25, 26]. The dataset contains drug sensitivity measurements for cancer cell lines screened against various compounds (see Table 1 for details).

| Metric                       | Value   |
|------------------------------|---------|
| Total Samples                | 177,310 |
| Unique Drugs                 | 208     |
| Unique Cell Lines            | 958     |
| Samples per Drug (mean)      | 852.5   |
| Samples per Drug (std)       | 305.1   |
| Samples per Cell Line (mean) | 185.1   |
| Samples per Cell Line (std)  | 35.9    |
| IC50 Mean                    | 2.0071  |
| IC50 Std                     | 2.7147  |
| IC50 Min                     | -9.9334 |
| IC50 Max                     | 12.3591 |
| Screening Coverage (%)       | 88.98   |

**Table 1.** Summary of the dataset statistics

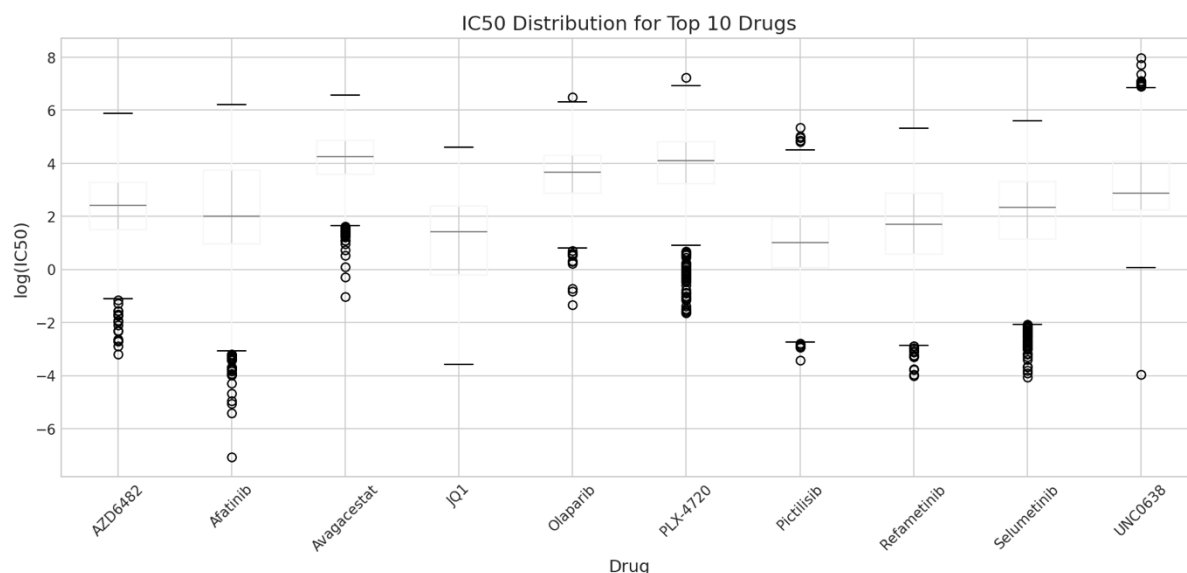
The dataset comprises 177,310 drug response measurements spanning 208 unique drugs and 958 unique cell lines. Each cell line is characterized by expression values for 17,737 genes. The log IC50 values range from negative 9.93 to 12.36, with a mean of 2.01 and standard deviation of 2.71.



**Figure 1.** Distribution of drug response values (log IC50) across the GDSC1 dataset. Left: Histogram showing the frequency distribution with mean (red dashed line, 2.01) and median (yellow dashed line, 2.46) indicated. Right: Box plot summarizing the distribution with outliers shown as individual points.

Figure 1 presents the overall distribution of drug response values in the dataset. The histogram reveals an approximately normal distribution of log IC50 values centered around the mean of 2.01 (red dashed line) with a median of 2.46 (yellow dashed line). The slight left skew, evidenced by the mean being lower than the median, indicates a tail of highly sensitive drug

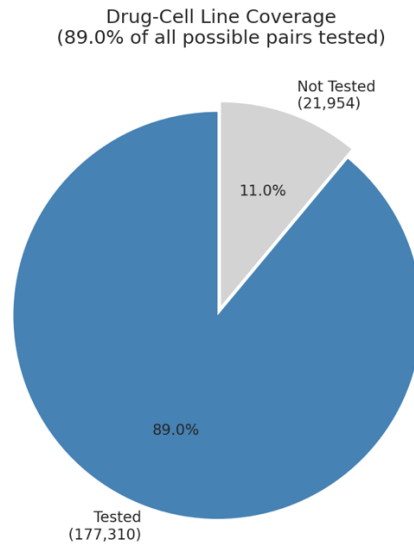
cell line pairs with very low IC50 values. The accompanying box plot highlights the presence of outliers at both extremes, representing exceptionally sensitive (log IC50 below negative 5) and resistant (log IC50 above 10) drug cell line pairs that pose challenges for accurate prediction.



**Figure 2.** IC50 Distribution for Top Drugs box plots: Drug-specific response variability

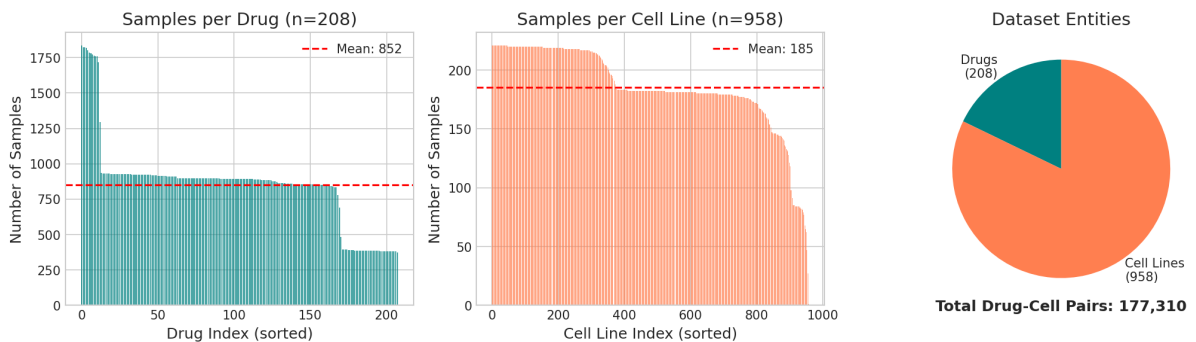
Figure 2 presents box plots of log IC50 distributions for the ten most frequently tested drugs in the dataset. Substantial variability exists both within and across drugs. Some compounds such as Afatinib and Pictilisib show relatively tight distributions centered at low IC50 values, indicating consistent potency across cell lines. Others like UNC0638 and Avagacestat exhibit wider spreads spanning over 10 log units, reflecting heterogeneous responses depending on cell line context. This drug-specific variability highlights the importance of cell line genomic features in predicting response and suggests that certain drugs may be inherently more challenging to model accurately.





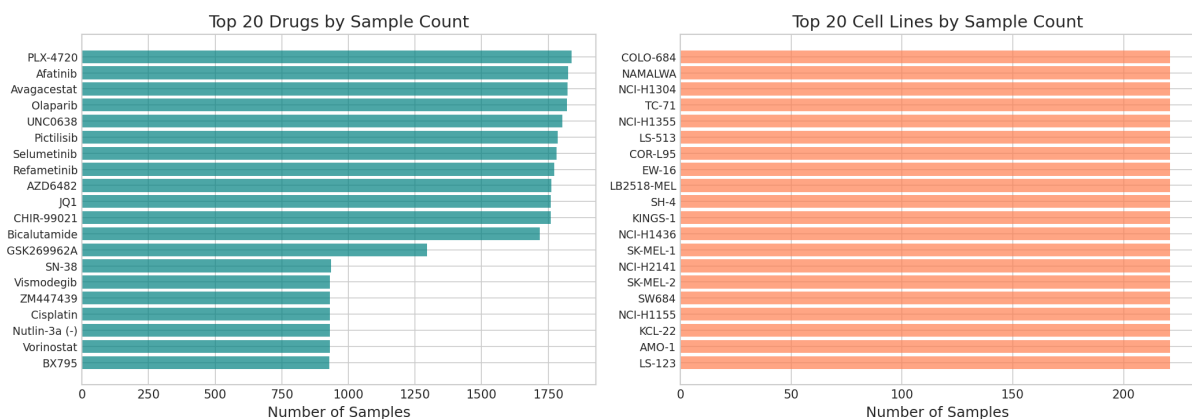
**Figure 4.** Drug-Cell Line Coverage pie chart: Shows 89% of possible pairs were tested

Figure 4 illustrates the coverage of the drug cell line screening matrix. Of the 199,264 possible drug cell line combinations ( $208 \text{ drugs} \times 958 \text{ cell lines}$ ), 177,310 pairs (89%) have been experimentally tested, leaving 21,954 pairs (11%) without measured response values. This high coverage rate reduces potential biases that could arise from systematic missingness and provides a robust foundation for model training and evaluation.



**Figure 5.** Samples per Drug and Cell Line Histograms: Distribution of screening coverage

Figure 5 displays the distribution of samples across drugs and cell lines. The left panel shows that each drug was tested against an average of 852 cell lines (standard deviation: 305), with most drugs screened against 800 to 950 cell lines, though some drugs have notably fewer measurements. The center panel reveals that each cell line was exposed to an average of 185 drugs (standard deviation: 36), with a more uniform distribution compared to the drug side. The pie chart confirms the dataset comprises 208 drugs and 958 cell lines, totalling 177,310 drug cell line pairs. This relatively balanced screening design helps ensure that learned representations generalize across both drug and cell line dimensions.



**Figure 6.** Top 20 Drugs and Cell Lines by Sample Count: Data availability across entities

Figure 6 ranks the most frequently represented entities in the dataset. The left panel shows that PLX-4720, Afatinib, and Avagacestat are the most extensively tested drugs, each appearing in over 1,800 samples, while compounds like BX795 have approximately half as many measurements. The right panel reveals that cell lines COLO-684, NAMALWA, and NCI-H1304 have the most comprehensive drug screening profiles, each tested against over 200 drugs. The relatively uniform distribution among top cell lines contrasts with the greater variability among drugs, indicating that the experimental design prioritized comprehensive cell line coverage. Models should be evaluated for potential performance variations between well-represented and underrepresented entities.

The data was partitioned into training, calibration, and test sets containing 124,117, 26,596, and 26,597 samples respectively. The calibration set is essential for conformal prediction, as it provides the held-out data used to compute the conformity score quantile. A roughly 70/15/15 split was employed to balance having sufficient data for model training while retaining adequate samples for calibration and evaluation.

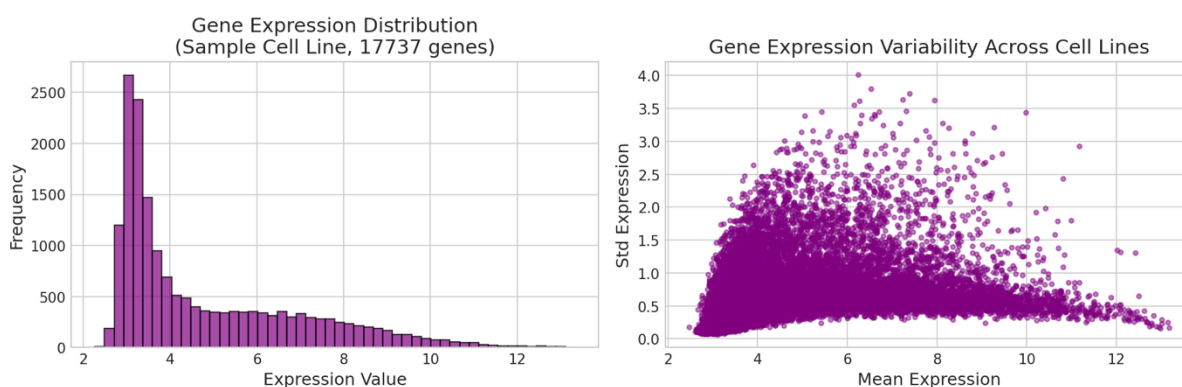
### 3.3 Model Architecture

The proposed model architecture consists of three main components: a drug encoder that processes molecular graphs, a cell line encoder that processes gene expression profiles, and a fusion network that combines these representations to produce the final prediction.

The drug encoder employs a three-layer Graph Convolutional Network with residual connections. Each GCN layer performs neighbourhood aggregation following the formulation of Kipf and Welling [27], where node representations are updated by aggregating features from neighbouring nodes weighted by the normalized adjacency matrix. The atom features include atomic number, degree, formal charge, hybridization state, aromaticity, and other chemical properties, resulting in a 69-dimensional input feature vector per atom. After message passing, a global mean pooling operation aggregates node features into a fixed length graph level representation.

The choice of Graph Convolutional Networks for drug encoding is motivated by their ability to learn hierarchical representations of molecular structure through iterative neighbourhood aggregation. In each GCN layer, atom representations are updated by aggregating information from bonded neighbours, allowing the network to capture increasingly complex substructural patterns as depth increases. The first layer captures immediate atomic environments such as functional groups, the second layer extends this to larger fragments, and the third layer enables the model to perceive molecule-wide structural motifs relevant to biological activity. Residual connections were incorporated between GCN layers to facilitate gradient flow during training and enable the network to preserve low-level atomic features alongside higher-order structural representations. The global mean pooling operation that follows the GCN layers computes a fixed-length molecular fingerprint by averaging all atom representations, ensuring permutation invariance with respect to atom ordering while retaining sensitivity to the overall molecular composition. This learned representation offers advantages over traditional fixed fingerprints such as ECFP or MACCS keys, as the model can adaptively identify structural features most predictive of drug response rather than relying on predefined substructure patterns.

The cell line encoder processes the 17,737-dimensional gene expression vector through a multilayer perceptron. Gene expression values were standardized to have zero mean and unit variance across the dataset.



**Figure 7.** Gene Expression: Characterization of cell line features

Figure 7 characterizes the gene expression features used to represent cell lines. The left panel shows the distribution of expression values for a representative cell line across all 17,737 genes, exhibiting a right-skewed distribution with most genes showing low to moderate expression and fewer genes highly expressed. The right panel displays the relationship between mean expression and expression variability (standard deviation) across cell lines for each gene. Genes with higher mean expression tend to exhibit greater variability, with a subset of highly variable genes in the upper right quadrant likely representing biologically informative features that distinguish cell line responses to drug treatment.

The MLP consists of fully connected layers with batch normalization and dropout for regularization, progressively reducing the dimensionality to match the drug embedding size.

The fusion network concatenates the drug and cell line embeddings and passes them through additional fully connected layers to produce the scalar IC50 prediction. Dropout with probability 0.2 was applied throughout the network to prevent overfitting and enable Monte Carlo dropout inference. The complete model contains 4,708,353 trainable parameters.

### 3.4 Training Procedure

The model was trained using the Adam optimizer with mean squared error loss. Training proceeded for a maximum of 100 epochs with early stopping based on validation loss, using a patience of 15 epochs. The batch size was set to 128. Model selection was performed by retaining the checkpoint with the lowest validation loss.

Drug molecules provided as SMILES strings were converted to molecular graphs using RDKit [28]. All 208 drugs were successfully converted without failures. Atom features were extracted including atomic number, chirality, degree, formal charge, number of hydrogens, hybridization, aromaticity, and ring membership. Edge features encoded bond type, stereochemistry, and conjugation.

### 3.5 Conformal Prediction

Split conformal prediction was employed to transform point predictions into prediction intervals with coverage guarantees. The procedure operates as follows. First, the trained model generates predictions on the calibration set. Then, conformity scores are computed as the absolute residuals between predictions and true values. After which, the  $(1 - \alpha) \times (1 + [1/n])$  quantile of the conformity scores is computed, where  $\alpha$  is the desired miscoverage rate and  $n$  is the calibration set size. In the end, prediction intervals for test points are constructed by adding and subtracting this quantile from the point predictions.

The key theoretical result underlying conformal prediction is that if the calibration and test data are exchangeable draws from some distribution  $P$ , then the prediction interval contains the true value with probability at least  $1 - \alpha$  [6]. Formally, for a new test point  $(x_{n+1}, y_{n+1})$ :

$$P(y_{n+1} \in \hat{C}(x_{n+1})) \geq 1 - \alpha$$

where  $\hat{C}(x_{n+1})$  denotes the prediction interval. This guarantee holds regardless of the true distribution  $P$  and requires no assumptions about model correctness.

For the primary experiments,  $\alpha$  was set to 0.1, corresponding to a target coverage of 90%. Additional experiments evaluated coverage at  $\alpha$  values of 0.2, 0.15, 0.05, and 0.02, corresponding to target coverages of 80%, 85%, 95%, and 98% respectively.

### **3.6 Baseline: Monte Carlo Dropout**

Monte Carlo dropout was implemented as a baseline uncertainty quantification method for comparison with conformal prediction. This approach, proposed by Gal and Ghahramani [19], interprets dropout at test time as approximate Bayesian inference.

The implementation performed 50 stochastic forward passes with dropout enabled at test time. For each test sample, the prediction was computed as the mean across all forward passes, and uncertainty was estimated as the standard deviation. Prediction intervals were then constructed as the mean plus or minus a multiple of the standard deviation.

This approach has the advantage of requiring no additional calibration data and providing input dependent uncertainty estimates that vary based on the specific drug cell line pair being evaluated. However, unlike conformal prediction, MC dropout does not provide formal coverage guarantees.

### **3.7 Evaluation Metrics**

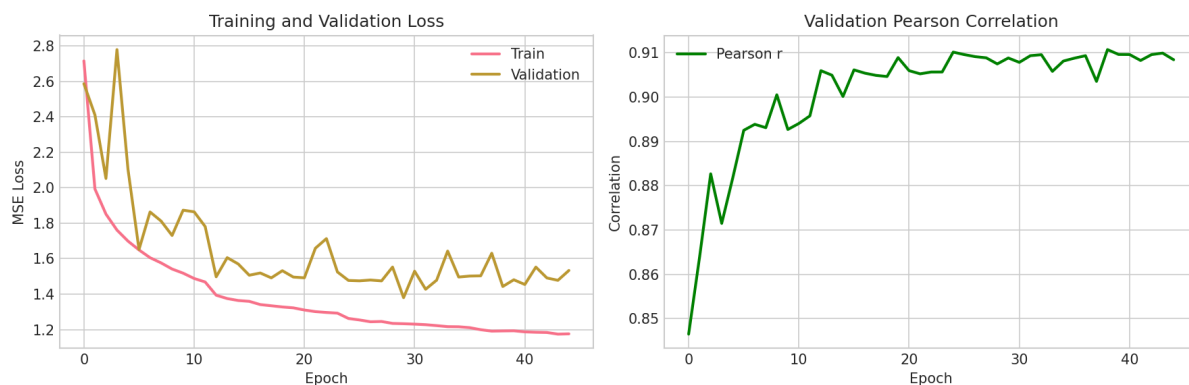
Model performance was evaluated using standard regression metrics including root mean squared error (RMSE), mean absolute error (MAE), Pearson correlation coefficient, and Spearman rank correlation coefficient.

For uncertainty quantification, the primary metrics were empirical coverage and mean interval width. Coverage is the proportion of test samples whose true values fall within the predicted intervals. The coverage gap measures the difference between empirical and target coverage. Mean interval width quantifies the average size of prediction intervals, with narrower intervals preferred given equal coverage. The uncertainty error correlation measures whether predicted uncertainty is associated with actual prediction errors, which indicates whether the model knows when it is likely to be wrong.

## **4. Results and Analysis**

### **4.1 Predictive Performance**

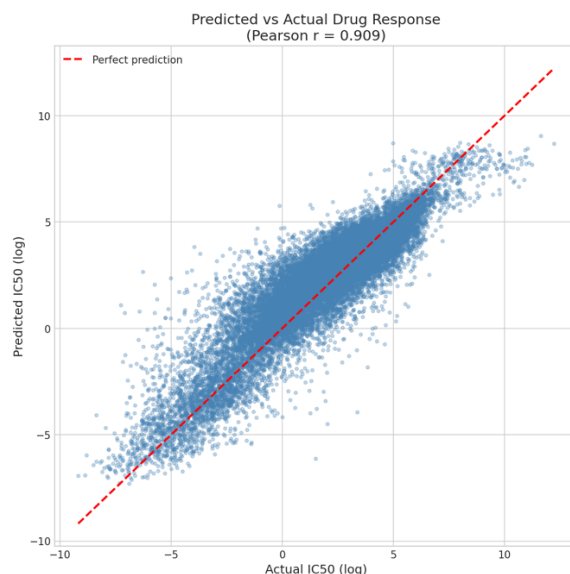
The trained GNN model achieved strong predictive performance on the held-out test set. The model attained an RMSE of 1.184 and MAE of 0.876 on the log IC50 scale. The Pearson correlation between predicted and actual values was 0.909, and the Spearman rank correlation was 0.889. These results indicate that the model captures both the magnitude and relative ordering of drug sensitivities with high fidelity.



**Figure 8.** Training and Validation Loss & Validation Pearson Correlation

Figure 8 displays the training dynamics over the course of optimization. The training loss decreased smoothly from an initial value of approximately 2.7 to below 1.2, while the validation loss exhibited more variance but followed a generally decreasing trend before early stopping triggered at epoch 45. The validation Pearson correlation increased rapidly during the first 10 epochs, rising from 0.85 to approximately 0.89, then continued to improve gradually, reaching 0.91 by the end of training. The gap between training and validation loss suggests some overfitting, though the regularization through dropout and early stopping prevented severe degradation of generalization performance.

The strong predictive performance achieved by the GCN-based model demonstrates the effectiveness of learning molecular representations directly from graph structure for drug response prediction. Unlike fingerprint-based approaches that encode molecules as fixed binary or count vectors, the GCN learns task-specific representations that emphasize structural features relevant to IC50 prediction. The message passing mechanism enables the model to capture both local chemical environments (such as hydrogen bond donors, aromatic systems, and charged groups) and global molecular properties (such as overall shape and lipophilicity) within a unified learned representation. The smooth convergence of training loss and the consistent improvement in validation correlation throughout training (Figure 8) suggest that the GCN successfully identified generalizable structure-activity relationships rather than memorizing training examples. The relatively small gap between training and validation performance further indicates that the combination of dropout regularization, early stopping, and the inductive bias of the graph convolution operation provided sufficient regularization to prevent overfitting despite the model's 4.7 million parameters.



**Figure 9.** Predicted vs Actual Drug Response scatter plot

Figure 9 shows the scatter plot of predicted versus actual log IC<sub>50</sub> values on the test set, achieving Pearson  $r = 0.909$ . The points cluster tightly around the diagonal line of perfect prediction across the entire range of responses, from highly sensitive cases with log IC<sub>50</sub> below negative 5 to resistant cases with log IC<sub>50</sub> above 10. The consistency of prediction quality across this wide range demonstrates that the model has learned meaningful representations applicable to diverse drug response scenarios.

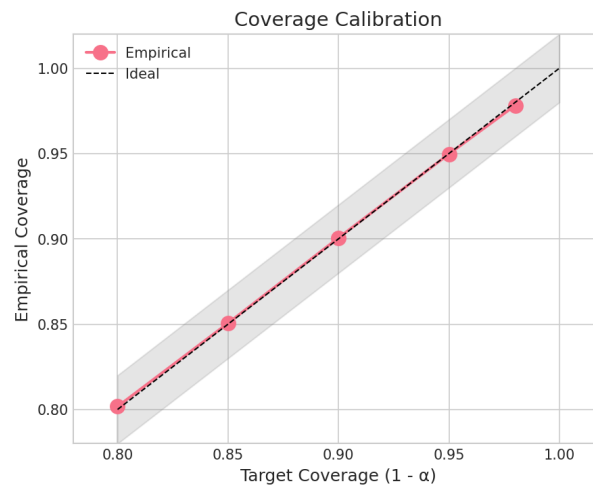
## 4.2 Conformal Prediction Coverage

The conformal prediction intervals achieved excellent calibration across multiple coverage levels. At the target coverage of 90%, the empirical coverage was 90.2%, resulting in a coverage gap of only 0.25 percentage points. A binomial test comparing the observed coverage to the target yielded a p value of 0.175, confirming that the empirical coverage is statistically consistent with the theoretical guarantee.

| alpha       | target_coverage | empirical_coverage | interval_width |
|-------------|-----------------|--------------------|----------------|
| <b>0.2</b>  | 0.80            | 0.8017             | 2.8044         |
| <b>0.15</b> | 0.85            | 0.8506             | 3.2331         |
| <b>0.1</b>  | 0.90            | 0.9025             | 3.7951         |
| <b>0.05</b> | 0.95            | 0.9497             | 4.8852         |
| <b>0.02</b> | 0.98            | 0.9782             | 6.2826         |

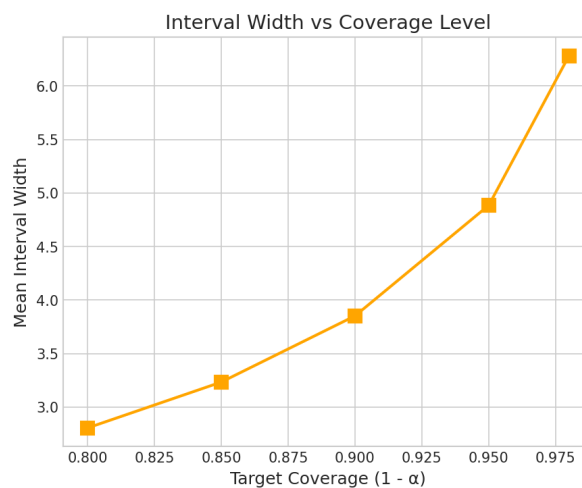
**Table 2.** Coverage analysis

Table 2 presents results across different coverage levels. At 80% target coverage, the empirical coverage was 80.2% with a mean interval width of 2.80. At 85% target, coverage was 85.1% with width 3.23. At 95% target, coverage was 95.0% with width 4.89. At 98% target, coverage was 97.8% with width 6.28. In all cases, the coverage guarantee was satisfied, with empirical coverage meeting or slightly exceeding the target.



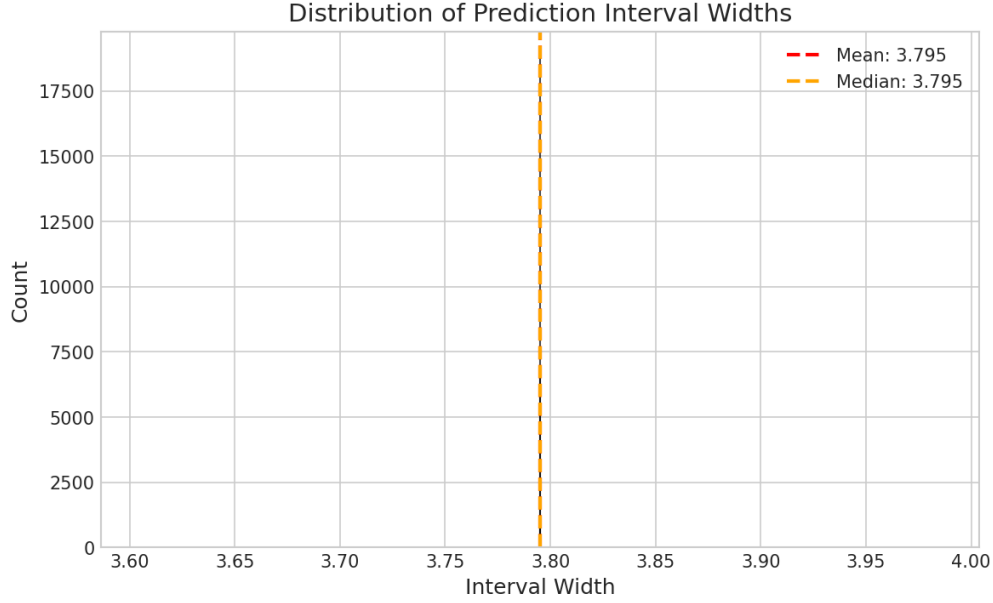
**Figure 10.** Coverage calibration plot

Figure 10 illustrates the relationship between target and empirical coverage. The points fall almost exactly on the diagonal line representing perfect calibration. The shaded region indicates the expected range of statistical fluctuation, and all empirical values fall within this band. This demonstrates that conformal prediction successfully translates the theoretical coverage guarantee into practice.



**Figure 11.** Interval Width vs Coverage Level; shows relationship between target coverage and mean prediction interval width

The coverage width trade-off exhibits the expected behaviour (coverage guarantee vs. interval precision), as shown in Figure 11.



**Figure 12.** Distribution of Prediction Interval Widths: Confirms constant width from split conformal (width = 3.795)

Figure 12 confirms a key characteristic of split conformal prediction - all prediction intervals have identical width. The distribution shows a single spike at 3.795 log IC50 units, reflecting the constant interval half-width of 1.898 (the calibrated quantile) applied symmetrically around each point prediction. This uniformity arises because standard split conformal prediction computes a single quantile from the calibration set residuals and applies it universally to all test predictions, regardless of input characteristics. While this approach guarantees marginal coverage, it cannot adapt to heteroscedastic errors where some predictions may warrant wider or narrower intervals.

Higher coverage levels require wider intervals, with the relationship being approximately linear for moderate coverage levels but accelerating for extreme coverage values. Moving from 80% to 90% coverage increases the mean width by approximately 1 unit, while moving from 90% to 98% increases width by nearly 2.5 units. This reflects the heavy tailed nature of the error distribution, where achieving very high coverage requires accommodating rare large errors.

### 4.3 Comparison with Monte Carlo Dropout

The comparison between conformal prediction and Monte Carlo dropout revealed substantial differences in coverage behaviour.

Monte Carlo dropout with 50 forward passes yielded a mean uncertainty of 0.403, measured as the standard deviation across stochastic predictions. However, the coverage achieved using dropout derived intervals was only 48.6%, dramatically undercovering relative to the 90% target that conformal prediction achieved. This severe undercoverage demonstrates that the uncertainty estimates from MC dropout do not directly translate into calibrated prediction

intervals. The uncertainty error correlation for MC dropout was 0.114, indicating a weak positive relationship between predicted uncertainty and actual prediction errors. While this suggests that MC dropout provides some signal about when predictions are unreliable, the correlation is too weak to be practically useful for identifying high error predictions.

In contrast, conformal prediction achieved 90.2% coverage with a mean interval width of 3.795. The coverage gap of only 0.25 percentage points demonstrates the reliability of the conformal guarantee. While conformal prediction produces constant width intervals that do not adapt to input dependent uncertainty, this simplicity comes with the crucial benefit of guaranteed coverage. Table 3 demonstrates an important property of conformal prediction; it functions as a post hoc wrapper that does not alter the underlying model's point predictions. The RMSE, MAE, Pearson correlation, and Spearman correlation remain identical whether or not conformal prediction is applied. Conformal prediction simply augments these point estimates with calibrated prediction intervals, adding coverage guarantees and interval width information without any computational overhead during model training.

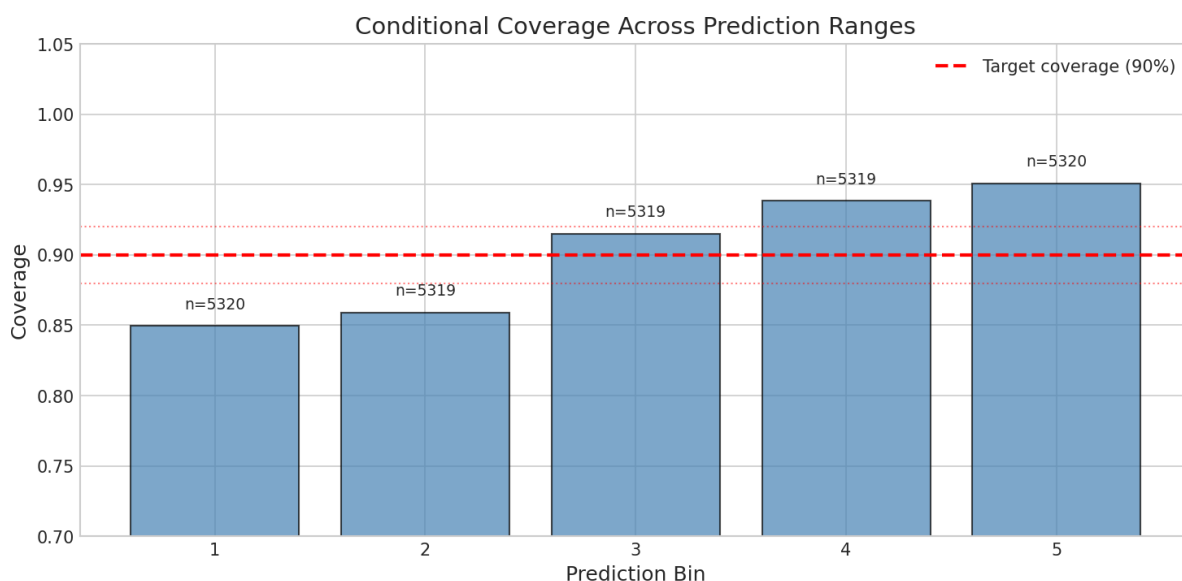
| Metric              | GNN (No UQ) | GNN + Conformal |
|---------------------|-------------|-----------------|
| RMSE                | 1.1844      | 1.1844          |
| MAE                 | 0.8756      | 0.8756          |
| Pearson r           | 0.9094      | 0.9094          |
| Spearman r          | 0.8886      | 0.8886          |
| Coverage (90%)      | N/A         | 90.2%           |
| Mean Interval Width | N/A         | 3.7951          |

**Table 3.** Comparison of GNN performance with and without conformal prediction, demonstrating that conformal prediction adds coverage guarantees (90.2%) and interval width (3.7951) without altering point prediction metrics

The fundamental difference between these approaches lies in their theoretical foundations. MC dropout provides a heuristic uncertainty estimate based on the variability of predictions under different dropout masks, but this variability is not calibrated to achieve any particular coverage level. Conformal prediction, by using a held out calibration set to determine the appropriate interval width, ensures that the resulting intervals contain the true value with the specified probability.

#### 4.4 Conditional Coverage Analysis

While conformal prediction guarantees marginal coverage, meaning that averaged across all test points the coverage meets the target, it does not guarantee conditional coverage for specific subgroups. Analyzing coverage conditional on the predicted value reveals heterogeneity in performance across different response ranges.



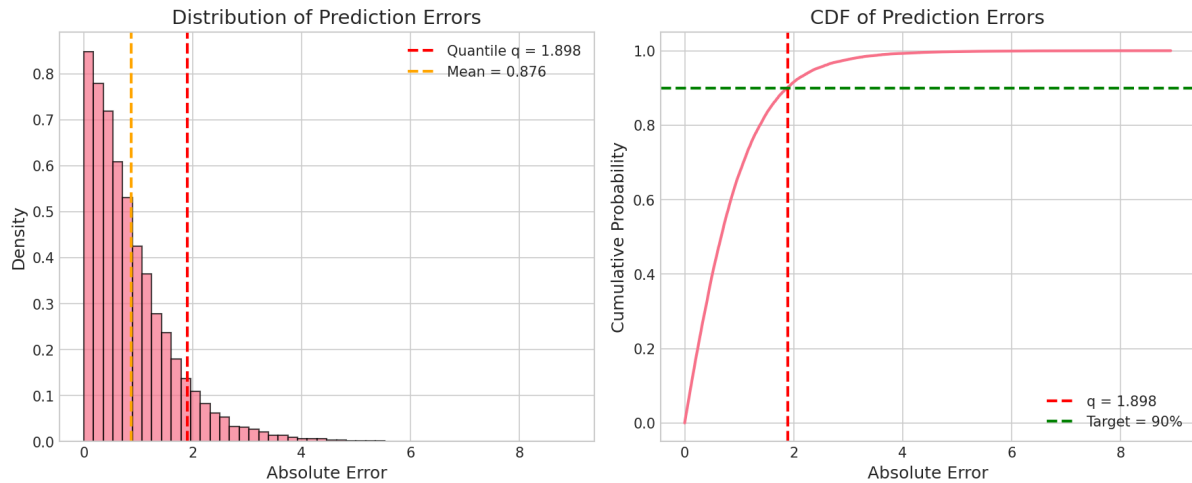
**Figure 13.** Conditional coverage analysis across prediction quintiles, revealing undercoverage for low predictions and overcoverage for high predictions relative to the 90% target

Figure 13 and the associated analysis show coverage broken down by prediction quintiles. For samples with predictions in the lowest bin, corresponding to log IC50 values below 0.80, the coverage was 84.9%, falling short of the 90% target. The second bin achieved 85.9% coverage. In contrast, the higher prediction bins achieved coverage exceeding the target: 91.5% for bin three, 93.8% for bin four, and 95.1% for bin five.

This pattern indicates that conformal prediction tends to undercover for sensitive drug cell line pairs and overcover for resistant pairs. The likely explanation relates to heteroscedasticity in the underlying errors. Predictions of low IC50 values may be inherently more uncertain due to the exponential nature of dose response curves, where small absolute changes in concentration translate to large changes on the log scale. The constant width intervals produced by standard conformal prediction cannot adapt to this variation, leading to the observed conditional coverage pattern.

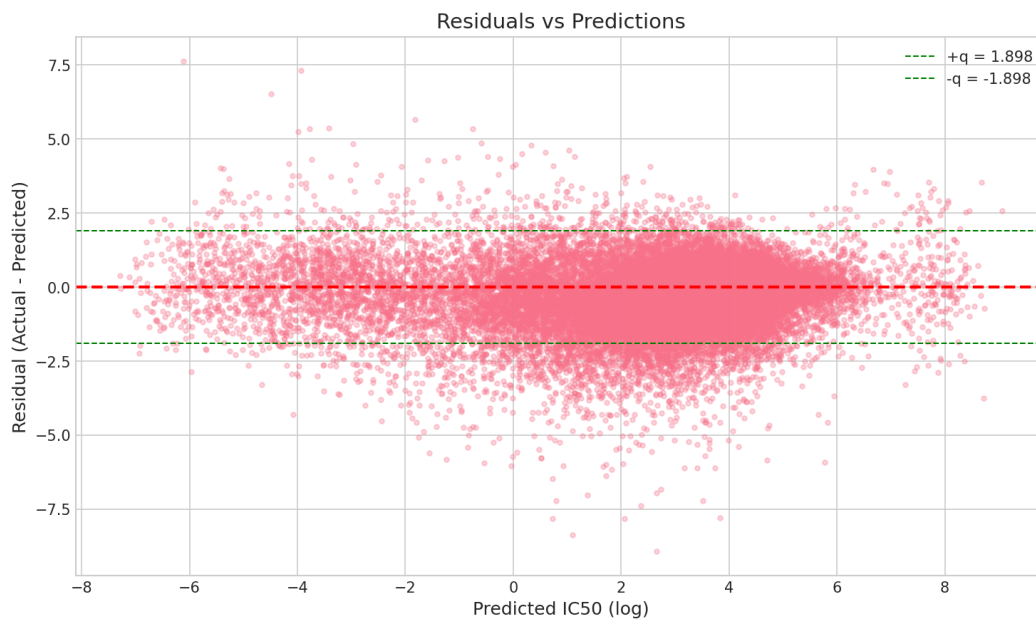
## 4.5 Error Analysis

The distribution of prediction errors provides insight into model behaviour. Figure 14 shows the histogram and cumulative distribution function of absolute errors. The distribution is right skewed, with a mean absolute error of 0.876 and median of 0.670, indicating that most predictions are quite accurate while a smaller fraction exhibit larger errors. The maximum absolute error observed was 8.92, representing cases where the model prediction was substantially incorrect.



**Figure 14.** Distribution of absolute prediction errors showing density histogram (left) and cumulative distribution function (right) with the calibrated quantile  $q = 1.898$  marked

The calibrated quantile of 1.898 corresponds to the 90th percentile of the error distribution on the calibration set. This value defines the half width of the 90% prediction intervals. The CDF plot confirms that approximately 90% of errors fall below this threshold, validating the conformal calibration procedure.



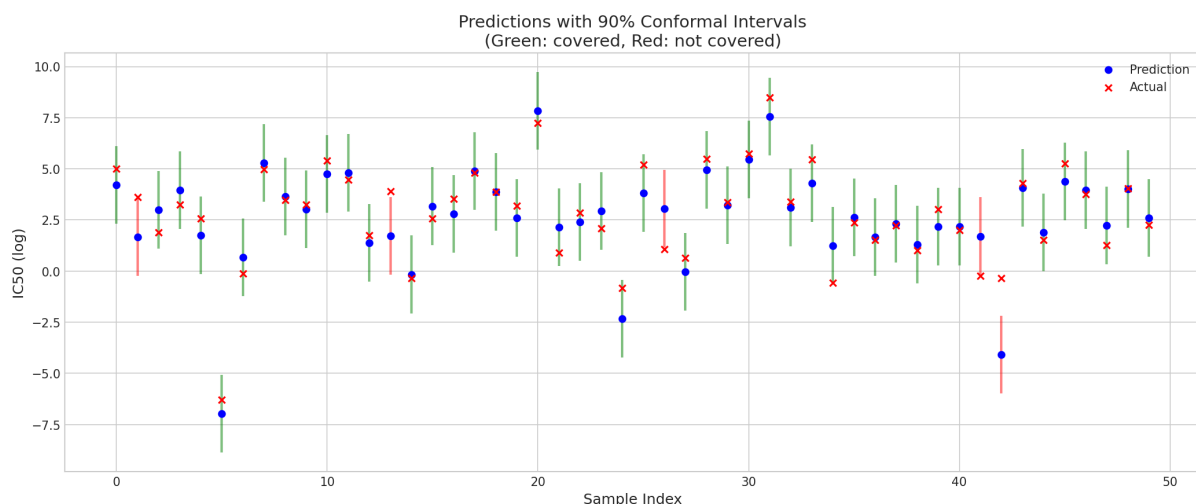
**Figure 15.** Residual plot showing prediction errors versus predicted values, with green dashed lines indicating the conformal quantile boundaries

The residual plot in Figure 15 displays the relationship between predicted values and prediction errors. The residuals are approximately centered around zero across the entire prediction range, indicating no systematic bias. The spread of residuals appears roughly constant, though with slightly more variance at extreme predicted values. The green lines indicate the positive and

negative quantile boundaries, and approximately 90% of points fall within these bounds as expected.

## 4.6 Prediction Interval Visualization

Figure 16 presents examples of prediction intervals for individual test samples. Each point shows the model prediction with the corresponding 90% conformal interval, along with the true IC50 value. Green intervals indicate cases where the true value falls within the interval, while red intervals indicate coverage failures.

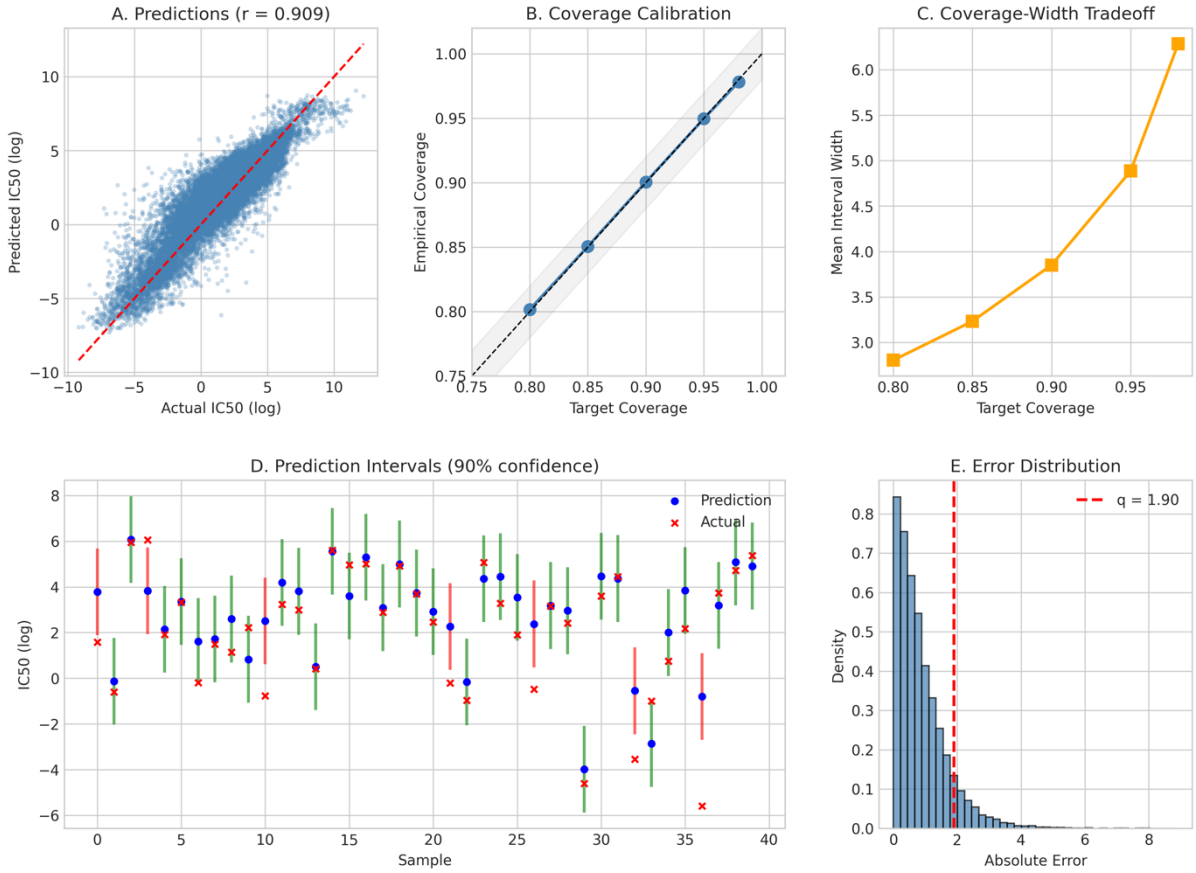


**Figure 16.** Example prediction intervals for test samples showing predictions (blue points), actual values (red crosses), and 90% conformal intervals

The visualization reveals that coverage failures occur throughout the prediction range but are more frequent for certain samples. In some cases, the true value falls just outside the interval boundary, representing near misses that would be covered with slightly wider intervals. In other cases, the true value is substantially outside the interval, indicating predictions where the model was confidently incorrect.

## 4.7 Results Summary

Figure 17 provides a comprehensive summary of the framework and results. Panel A shows the prediction scatter plot demonstrating high correlation. Panel B displays the coverage calibration curve. Panel C illustrates the coverage width tradeoff. Panel D shows example prediction intervals. Panel E presents the error distribution with the calibrated quantile marked.



**Figure 17.** Comprehensive summary of the framework showing (A) prediction scatter plot, (B) coverage calibration, (C) coverage-width tradeoff, (D) example prediction intervals, and (E) error distribution

The juxtaposition of panels B and C reveals the core tradeoff in conformal prediction; while coverage guarantees are reliably met across all confidence levels (Panel B), the cost of moving from 90% to 98% coverage is disproportionately high, nearly doubling the interval width (Panel C), suggesting that practitioners should carefully consider whether marginal coverage gains justify substantially reduced precision. Panel D visually illustrates the exchangeability assumption in action, where coverage failures (red intervals) appear randomly distributed across the prediction range rather than clustering in specific regions, providing empirical support for the validity of the conformal framework on this dataset. The right-skewed error distribution in Panel E, combined with the calibrated quantile of 1.90, explains why the constant-width intervals must be relatively wide; accommodating the long tail of occasional large errors is necessary to achieve the coverage guarantee, highlighting an inherent limitation when the underlying model produces heteroscedastic errors.

## 5. Discussion

### 5.1 Key Findings

This work demonstrates that conformal prediction provides a principled and effective approach to uncertainty quantification for GNN based drug response prediction.

To begin with, the combination of GNNs for drug encoding and MLPs for cell line encoding achieves strong predictive performance on the GDSC1 dataset. The Pearson correlation of 0.909 indicates that the model explains approximately 83% of the variance in drug response, which is notable given the complexity and heterogeneity of the biological system being modelled.

Secondly, split conformal prediction successfully provides coverage guarantees that hold in practice. Across five different coverage levels ranging from 80% to 98%, the empirical coverage consistently matched the target within statistical tolerance. This demonstrates that the exchangeability assumption underlying conformal prediction is reasonable for this application.

In addition, conformal prediction substantially outperforms Monte Carlo dropout in terms of coverage calibration. While MC dropout achieved only 48.6% coverage, conformal prediction achieved the target 90.2% coverage. This dramatic difference highlights the importance of using methods with formal guarantees rather than heuristic uncertainty estimates.

Furthermore, conditional coverage analysis reveals limitations of the standard conformal approach. The undercoverage for sensitive predictions and overcoverage for resistant predictions suggests that adaptive conformal methods, which adjust interval width based on predicted uncertainty, may be beneficial for this application.

From a deep learning perspective, this work signifies that graph neural networks provide an effective inductive bias for molecular property prediction tasks. The GCN architecture encodes the fundamental principle that molecular properties arise from the spatial arrangement of atoms and bonds, allowing the model to generalize across structurally diverse compounds. The three-layer architecture strikes a balance between expressiveness and oversmoothing; shallower networks would limit the receptive field to immediate atomic neighbourhoods, while deeper networks risk collapsing atom representations toward indistinguishable values. The fusion of graph-level drug representations with cell line gene expression vectors through concatenation followed by fully connected layers enables the model to learn complex, nonlinear interactions between drug structure and genomic context. This multimodal architecture reflects the biological reality that drug response depends on both the compound's mechanism of action (encoded in its structure) and the cellular context (encoded in gene expression), and the learned joint representation captures these dependencies in a data-driven manner.

## 5.2 Practical Implications

The framework developed in this work has several practical implications for drug discovery and precision oncology. The ability to provide calibrated prediction intervals enables more informed decision making in downstream applications. For virtual screening, compounds can be prioritized not only by their predicted potency but also by the confidence of those predictions. Compounds with narrow prediction intervals indicating high confidence may be prioritized for experimental validation, while those with wide intervals may require additional investigation.

In precision oncology applications, the prediction intervals provide clinicians with quantitative measures of uncertainty that can inform treatment decisions. A drug predicted to be highly effective with a narrow confidence interval represents a different clinical situation than one predicted to be effective but with substantial uncertainty. The coverage guarantees ensure that these intervals can be trusted to contain the true sensitivity with the stated probability.

The GCN-based approach offers practical advantages for drug discovery workflows beyond prediction accuracy. Because the model learns molecular representations end-to-end, it can be readily fine-tuned for related tasks such as predicting response to new drug classes or in different cancer types with minimal feature engineering. The learned drug embeddings could be extracted and used for downstream analyses such as clustering compounds by predicted mechanism of action or identifying structural analogs with similar predicted response profiles. Also, the attention weights or gradient-based attribution methods could be applied to the trained GCN to identify which molecular substructures most strongly influence predicted sensitivity, potentially offering interpretable insights into structure-activity relationships that complement traditional medicinal chemistry approaches.

## 5.3 Novelty and Contributions

Several aspects of this work represent novel contributions to the field. The integration of GNN based drug response prediction with conformal prediction has not been extensively studied in prior work. While conformal prediction has been applied to QSAR models, comprehensive evaluations using modern graph neural network architectures and large-scale drug sensitivity datasets are lacking.

The comparative analysis with Monte Carlo dropout provides valuable insights into the relative merits of different uncertainty quantification approaches. The finding that MC dropout substantially undercovered with only 48.6% coverage while conformal prediction achieved the target 90.2% highlights the importance of methods with formal guarantees.

The conditional coverage analysis identifies a previously unreported pattern of heterogeneous coverage across prediction ranges. This finding has implications for the development of adaptive conformal methods and suggests directions for methodological improvement.

## 6. Challenges and Limitations

Several challenges were encountered during the development and evaluation of this framework. The computational cost of training graph neural networks on molecular data is substantial, requiring GPU acceleration for practical training times. MC dropout inference with 50 forward passes also increases inference time compared to a single forward pass.

The constant width intervals produced by split conformal prediction represent a limitation for applications where uncertainty varies systematically across the input space. The conditional coverage analysis demonstrates that this limitation manifests in practice, with coverage varying across prediction ranges. Conformalized quantile regression or locally adaptive conformal methods could address this limitation but were not evaluated in this work.

The reliance on a held-out calibration set reduces the data available for training. With approximately 15% of data reserved for calibration, the model sees fewer training examples than would otherwise be possible. Cross conformal or jackknife+ approaches could potentially mitigate this limitation by reusing data across folds.

The dataset, while large by historical standards, represents a specific set of cell lines and compounds. The coverage guarantees of conformal prediction assume exchangeability between calibration and test data, which may not hold under distribution shift. Deploying this model on cell lines with substantially different characteristics or novel drug scaffolds may result in degraded coverage.

The interpretation of prediction intervals on the log IC50 scale requires care. An interval of width 3.8 on the log scale corresponds to a factor of approximately 45 on the original concentration scale. For some applications, this uncertainty may be too large to be practically useful, even though it accurately reflects the model's predictive limitations.

The weak uncertainty error correlation observed for MC dropout (0.114) suggests that neither approach provides strong signals about when individual predictions are likely to be incorrect. While conformal prediction guarantees marginal coverage, it does not identify which specific predictions will be errors.

Some limitations specific to the GCN architecture merit consideration. The standard GCN formulation operates on 2D molecular graphs and does not explicitly encode 3D conformational information, which can be critical for understanding binding interactions with protein targets. Incorporating 3D geometric information through equivariant graph neural networks or distance-aware message passing could potentially improve predictions for drugs whose activity is conformation-dependent. Moreover, the message passing paradigm in GCNs treats all edges (bonds) with equal importance during neighbourhood aggregation, though edge features encoding bond type were included in the implementation. More sophisticated architectures such as Graph Attention Networks could learn to weight different bonds according to their relevance for the prediction task. Additionally, the global mean pooling

operation, while ensuring permutation invariance, may lose information about the relative positions of important functional groups within the molecule. Hierarchical pooling methods or attention-based readout functions could preserve more structural detail in the final molecular representation. Importantly, GCNs are known to suffer from oversmoothing when many layers are stacked, limiting the effective receptive field; while the three-layer architecture used here avoids this issue, it may also limit the model's ability to capture very long-range intramolecular interactions.

Training deep learning models on drug response data presents extra challenges related to data heterogeneity and batch effects. The gene expression features used to represent cell lines originate from different experimental platforms and processing pipelines, potentially introducing technical variation that the model may inadvertently learn. While standardization was applied to mitigate this issue, more sophisticated batch correction methods or domain adaptation techniques could further improve robustness. The training procedure required careful tuning of hyperparameters including learning rate, dropout rate, and the balance between drug encoder and cell line encoder capacity. The computational cost of training, requiring GPU acceleration and approximately 45 epochs to converge, may limit rapid iteration during model development. Besides, the use of early stopping based on validation loss, while preventing overfitting, may result in suboptimal solutions if the validation set is not fully representative of the test distribution.

## 7. Conclusion and Future Work

This work presented a framework combining graph neural networks with conformal prediction for uncertainty aware cancer drug response prediction. The approach achieves strong predictive performance with a Pearson correlation of 0.909 while providing prediction intervals with rigorous coverage guarantees. Empirical evaluation confirmed that 90% prediction intervals achieve 90.2% coverage on held-out test data, validating the theoretical properties of conformal prediction.

Comparative analysis demonstrated clear advantages of conformal prediction over Monte Carlo dropout for generating calibrated prediction intervals. MC dropout achieved only 48.6% coverage, substantially below the 90% target, while conformal prediction achieved the guarantee by construction. This finding underscores the importance of using uncertainty quantification methods with formal coverage guarantees for critical applications.

Many directions for future work emerge from this study. Adaptive conformal methods that adjust interval width based on input dependent uncertainty estimates could address the conditional coverage limitations identified in this work. Conformalized quantile regression, which learns to predict different quantiles of the response distribution, represents one promising approach.

Extending the framework to handle distribution shift would enhance its practical applicability. Conformal prediction under covariate shift has been studied theoretically, and incorporating these methods could provide coverage guarantees when deploying models on new cell lines or drug classes.

Incorporating additional data modalities, such as drug target information, pathway annotations, or multi omics profiles, could improve both predictive performance and uncertainty estimation. The modular architecture developed in this work readily accommodates such extensions.

Finally, prospective validation through collaboration with experimental groups would provide the ultimate test of the framework's utility. Selecting compounds for experimental testing based on model predictions and confidence intervals, then measuring the actual hit rate, would demonstrate whether the uncertainty quantification translates to practical value in drug discovery workflows.

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